

Sharvesh Kandavel

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EDUCATION

BASc, Mechatronics Engineering | Presidential Scholarship Recipient

2025 – 2030

University of Waterloo

Waterloo, ON

- MTE 140 – Algorithms & Data Structures (C++) | MTE 121 – Digital Computation (C++ & Vex Robotics)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL (PostgreSQL, SQLite), HTML

Frameworks & Web: React, React Native, FastAPI, RESTful APIs, Microservices

AI/ML Architecture: RAG Pipelines, LLM Integration, Python Data Modeling, Data Retrieval

Tools & Workflows: Selenium, Jira, Agile/Scrum, GitHub, Unix/Linux CLI, Cloud Platforms (GCP)

EXPERIENCE

Technical Founder

2026 – Present

Credvan

- Building responsive, scalable single-page web applications (SPAs) for employer portals using **React** and **TypeScript**
- Designing Python risk-scoring models to enforce strict transaction limits and withdrawal caps
- Architecting backend web services using **FastAPI REST endpoints** and Convex to process dynamic JSON data payloads
- Engineered and deployed credvan.com utilizing AI-assisted CLI workflows to accelerate prototyping
- Building an **ETL pipeline** mapping external REST APIs (ADP, Gusto, Workday) to a unified, relational database schema
- Ensured Plaid + Stripe EWA architecture complies strictly with the 2025 CFPB Advisory Opinion

Applied AI Program

Aug 2025

IIT Madras

Chennai, India

- Built a context-aware AI chatbot connected to a document knowledge base for retrieval-augmented Q&A
- Implemented **RAG pipelines** using AnythingLLM with vector embeddings for document ingestion and aware retrieval
- Deployed and benchmarked local machine learning models via **LM Studio**, strictly evaluating processing latency and prompt engineering trade-offs

Web Developer

May 2026 – Aug 2026

WE Accelerate Program | WEA Employer-Funded Recipient

- Executed structural SEO optimizations across two independent websites, increasing performance metrics by **45%** and **40%**
- Configured Google Analytics and authored technical documentation to train the employer on long-term data utilization
- Awarded the **WEA Employer-Funded Award** for outstanding contribution to the program

PROJECTS

🔗 Revamp | React, TypeScript, Three.js, FastAPI, PostgreSQL

- Developed a full-stack PC customization platform with **TypeScript** dashboards to curate tailored hardware configurations
- Automated data extraction via external **REST APIs (Rainforest API)**, dynamically live component specifications
- Engineered interactive 3D PC hardware models using **Three.js** and React Three Fiber for high-performance rendering
- Managed complex **JSONB** structures in a PostgreSQL database to allow for flexible metadata and real-time CRUD sync
- Enforced strict data isolation via **Row Level Security (RLS)** in PostgreSQL – sellers access only their own inventory

🔗 Bitgold | React, TypeScript, Convex, Clerk, Tailwind CSS

- Built a fintech platform investing spare change into fractional 24K gold, integrating a **Live Gold API** for real-time price
- Engineered a serverless **Convex** backend with TypeScript to execute real-time transactions via zero-polling WebSockets
- Implemented secure multi-tenant authentication and double-entry bookkeeping ledgers utilizing **Clerk** and JWTs
- Built responsive client interfaces in **React 19**, integrating interactive financial charts via Recharts and Framer Motion
- Designed strict TypeScript database schemas to securely manage fiat deposits, gold balances, and automated SIP schedules

🔗 Martial-Arts Motion Sensor | C++, Object-Oriented Design, Arduino Nano, MPU-6050 IMU

- Engineered a wearable tracking system on an Arduino Nano to capture and analyze martial arts swing mechanics
- Wrote robust algorithms to securely stream telemetry via Bluetooth with <100 ms latency for external visualization
- Applied **object-oriented C++** converting IMU data to swing scores – $\pm 10\%$ error; flagged 15–20% slower swings
- Integrated an MPU-6050 gyroscope to capture multi-axis motion data at a **200 Hz** sampling rate

🔗 Autonomous Sorting Robot | C++, VEX Robotics, PID Control, SolidWorks

- Designed and built an autonomous robot utilizing 3 motors and 4 sensors to detect, navigate, and sort objects by color
- Implemented dynamic **PID controllers** for precise mechanical positioning, achieving ± 1 cm accuracy
- 3D-modeled and printed custom components and containers using SolidWorks to optimize fit and functionality